

Smart Metro Ticket Booking System

Requirement Analysis Phase

Team ID

Project Name: Smart Metro Ticket Booking System

Maximum Marks: 4 Marks

2. Data Flow Diagram & User Stories

This document lays out the system's data flow at Level 0 (context) and Level 1 (process breakdown), then presents user stories capturing what each persona needs functionally: Passengers, Station Managers, the Metro Operations Team, and IT Admins.

2.1 Data Flow Diagram — Level 0 (Context)

Figure 2.1 — External entities that exchange data with the Metro Ticket Booking System

The context diagram identifies four human actors and two external systems exchanging data with the core booking platform: Passengers kick off booking requests; Station Managers keep station and fare master data current; the Metro Operations Team consumes operational reports; IT Admins handle configuration and access; the Payment Gateway processes transactions; and the Notification Channel delivers tickets and alerts.

2.2 Data Flow Diagram — Level 1 (Process Decomposition)

Figure 2.2 — Breakdown of the booking system into its six core processes

The system splits into six processes: Station & Route Selection, Fare Calculation, Payment Processing, QR Ticket Generation, Notification Dispatch, and Reporting & Analytics. These map directly onto Flow Designer sub-flows and Script Includes, reading from and writing to three core data stores — Station Master, Fare Rule Table, and Ticket/QR Records.

2.3 User Stories

User stories are organised by persona and follow the standard "As a <role>, I want <capability>, so that <benefit>" format, each backed by acceptance criteria.

Passenger (End User)

ID	User Story	Acceptance Criteria
US-01	As a passenger, I want to pick my source and destination stations from a searchable catalog form, so that I can book without a counter visit.	Catalog form loads the station list from Station Master; invalid source/destination pairs are blocked; search-as-you-type is supported.
US-02	As a passenger, I want to see the calculated fare right after entering travel details, so that I know the exact amount before paying.	Fare updates live as station, passenger type, or ticket count changes; Student/Senior Citizen discount applies automatically.
US-03	As a passenger, I want to pay digitally via UPI, card, or wallet, so that exact cash isn't required.	Payment sub-flow supports at least three digital modes; failed payments offer a clear retry path.
US-04	As a passenger, I want my ticket as a QR code right after payment, so that I can travel without a printed ticket.	QR is generated within seconds of payment confirmation and delivered via app notification and email/WhatsApp.
US-05	As a passenger, I want to view and re-download past bookings, so that I have a record for reimbursement or reuse.	Booking history is available in the Service Portal, filterable by date and route.

Station Manager

ID	User Story	Acceptance Criteria
US-06	As a station manager, I want to maintain station details and fare zones, so that booking data stays accurate.	Station Master records are editable through a controlled catalog/table with an approval workflow for changes.
US-07	As a station manager, I want to validate QR tickets at the gate in real time, so that only paid passengers get through.	QR scan returns valid/invalid/used status within 2 seconds; duplicate scans are flagged.
US-08	As a station manager, I want a dashboard of today's ticket volume at my station, so that I can plan staffing and manage crowding.	Dashboard refreshes in near real time and is filterable by hour.

Metro Operations Team

ID	User Story	Acceptance Criteria
US-09	As a metro operations manager, I want consolidated analytics on peak travel times and popular routes, so that I can plan schedules and resources.	Reports/Performance Analytics dashboards aggregate booking data by route, time band, and passenger type.
US-10	As a metro operations manager, I want visibility into digital-versus-cash revenue, so that I can track digital adoption.	Revenue split report is available monthly and exportable.
US-11	As a metro operations manager, I want automated alerts for failed or disputed transactions, so that I can act proactively.	Flow Designer raises an incident/case record for any payment exception.

IT Admin

ID	User Story	Acceptance Criteria
US-12	As an IT admin, I want to configure catalog variables and fare rules without code changes, so that business changes ship quickly.	Fare Rule Table and catalog variables are admin-configurable via the ServiceNow UI.
US-13	As an IT admin, I want role-based access across passenger, station manager, and operations views, so that data stays visible only to authorized users.	ACLs restrict table and UI Page access strictly by assigned role.
US-14	As an IT admin, I want monitoring and logs for the QR and payment integrations, so that I can troubleshoot failures quickly.	Integration Hub spokes log request/response with correlation IDs for traceability.